

MSDM5004

Numerical Methods and Modeling in Science

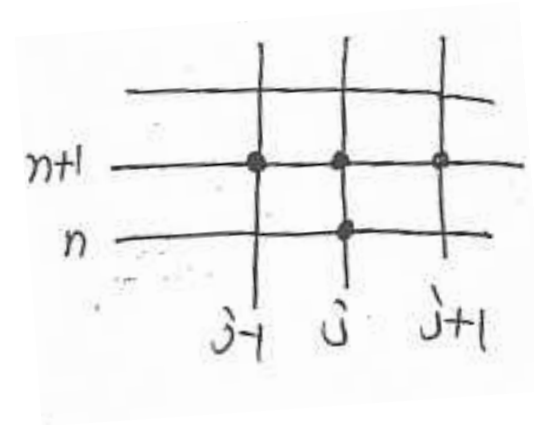
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Lecture 10

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1.5. An implicit scheme



$$\frac{U_j^{n+1} - U_j^n}{\Delta t} = \frac{U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}}{(\Delta x)^2}$$

$$-\mu U_{j-1}^{n+1} + (1 + 2\mu)U_j^{n+1} - \mu U_{j+1}^{n+1} = U_j^n$$

$$\mu = \frac{\Delta t}{(\Delta x)^2}$$

$$j = 1, 2, \dots, J-1.$$

a system of $J - 1$ linear equations in the $J - 1$ unknowns U_j^{n+1}

$$j = 1, 2, \dots, J-1.$$

Solving linear system

$$\begin{bmatrix} 1+2\mu & -\mu & & & \\ -\mu & 1+2\mu & -\mu & & \\ & \ddots & \ddots & \ddots & \\ & & -\mu & 1+2\mu & -\mu \\ & & & -\mu & 1+2\mu \end{bmatrix} \begin{bmatrix} U_1^{n+1} \\ U_2^{n+1} \\ \vdots \\ U_{J-2}^{n+1} \\ U_{J-1}^{n+1} \end{bmatrix} = \begin{bmatrix} U_1^n \\ U_2^n \\ \vdots \\ U_{J-2}^n \\ U_{J-1}^n \end{bmatrix}$$

- Tridiagonal system.
- Can be solved efficiently by Thomas algorithm with $O(N)$ operations. Here $N=J-1$.

Truncation error

$$T(x_j, t_n) = \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} - \frac{u(x_{j+1}, t_{n+1}) - 2u(x_j, t_{n+1}) + u(x_{j-1}, t_{n+1}))}{(\Delta x)^2}$$

Taylor expansion at (x_j, t_{n+1}) :

$$u(x_j, t_n) = u(x_j, t_{n+1}) - \Delta t \frac{\partial u}{\partial t}(x_j, t_{n+1}) + \frac{1}{2}(\Delta t)^2 \frac{\partial^2 u}{\partial t^2}(x_j, t_{n+1}) + O((\Delta t)^3).$$

$$\frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} = \frac{\partial u}{\partial t}(x_j, t_{n+1}) - \frac{1}{2}\Delta t \frac{\partial^2 u}{\partial t^2}(x_j, t_{n+1}) + O((\Delta t)^2).$$

Similarly,

$$\frac{u(x_{j+1}, t_{n+1}) - 2u(x_j, t_{n+1}) + u(x_{j-1}, t_{n+1}))}{(\Delta x)^2} = \frac{\partial^2 u}{\partial x^2}(x_j, t_{n+1}) + \frac{1}{12}(\Delta x)^2 \frac{\partial^4 u}{\partial x^4}(x_j, t_{n+1}) + O((\Delta x)^3).$$

Thus

$$\begin{aligned} T(x_j, t_n) = & \frac{\partial u}{\partial t}(x_j, t_{n+1}) - \frac{\partial^2 u}{\partial x^2}(x_j, t_{n+1}) \\ & - \frac{1}{2} \Delta t \frac{\partial^2 u}{\partial t^2}(x_j, t_{n+1}) - \frac{1}{12} (\Delta x)^2 \frac{\partial^4 u}{\partial x^4}(x_j, t_{n+1}) \\ & + O((\Delta t)^2) + O((\Delta x)^3) \end{aligned}$$

From the PDE,

$$\frac{\partial u}{\partial t}(x_j, t_{n+1}) - \frac{\partial^2 u}{\partial x^2}(x_j, t_{n+1}) = 0.$$

Thus we have

$$\begin{aligned} T(x_j, t_n) = & - \frac{1}{2} \Delta t \frac{\partial^2 u}{\partial t^2}(x_j, t_{n+1}) - \frac{1}{12} (\Delta x)^2 \frac{\partial^4 u}{\partial x^4}(x_j, t_{n+1}) \\ & + O((\Delta t)^2) + O((\Delta x)^3) \end{aligned}$$

This means that the implicit scheme is also first order in t and second order in x .

Stability

Let

$$U_j^n = [\lambda(k)]^n e^{ik(j\Delta x)}.$$

Substitute it into the scheme,

$$\begin{aligned} & -\mu[\lambda(k)]^{n+1}e^{ik((j-1)\Delta x)} + (1 + 2\mu)[\lambda(k)]^{n+1}e^{ik(j\Delta x)} - \mu[\lambda(k)]^{n+1}e^{ik((j+1)\Delta x)} \\ & = [\lambda(k)]^n e^{ik(j\Delta x)}. \end{aligned}$$

Divide it by $[\lambda(k)]^n e^{ik(j\Delta x)}$,

$$-\mu\lambda(k)e^{-ik\Delta x} + (1 + 2\mu)\lambda(k) - \mu\lambda(k)e^{ik\Delta x} = 1.$$

$$\lambda(k)[1 + 2\mu - \mu(e^{-ik\Delta x} + e^{ik\Delta x})] = 1.$$

$$\lambda(k)(1 + 2\mu - 2\mu \cos k\Delta x) = 1.$$

$$\lambda(k) \left(1 + 4\mu \sin^2 \frac{k\Delta x}{2} \right) = 1.$$

$$\lambda(k) = \frac{1}{1 + 4\mu \sin^2 \frac{k\Delta x}{2}}.$$

We have

$$|\lambda(k)| \leq 1, \text{ for all } k.$$

$$|\lambda(k)|^n \leq 1, \text{ for all } k.$$

Thus the implicit scheme is unconditionally stable.

1.6. Weighted average method (θ -method)

An explicit method:
$$\frac{U_j^{n+1} - U_j^n}{\Delta t} = \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{(\Delta x)^2} \quad (1)$$

An implicit method:
$$\frac{U_j^{n+1} - U_j^n}{\Delta t} = \frac{U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}}{(\Delta x)^2} \quad (2)$$

Average of (1) and (2): $0 \leq \theta \leq 1$, $(1)*(1 - \theta) + (2)*\theta$

Weighted average method (θ -method):

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} = (1 - \theta) \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{(\Delta x)^2} + \theta \frac{U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}}{(\Delta x)^2} \quad (3)$$

or

$$-\theta\mu U_{j-1}^{n+1} + (1 + 2\theta\mu)U_j^{n+1} - \theta\mu U_{j+1}^{n+1} = (1 - \theta)\mu U_{j-1}^n + [1 - 2(1 - \theta)\theta\mu]U_j^n + (1 - \theta)\mu U_{j+1}^n. \quad (4)$$

$\theta = 0$, it is (1).

$\theta = 1$, it is (2)

$\theta = \frac{1}{2}$, it is

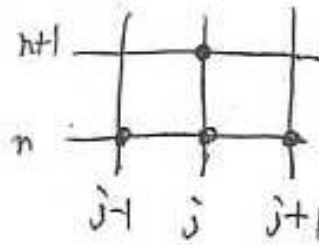
$$-\frac{\mu}{2}U_{j-1}^{n+1} + (1 + \mu)U_j^{n+1} - \frac{\mu}{2}U_{j+1}^{n+1} = \frac{\mu}{2}U_{j-1}^n + (1 - \mu)U_j^n + \frac{\mu}{2}U_{j+1}^n. \quad (5)$$

Crank-Nicolson method.

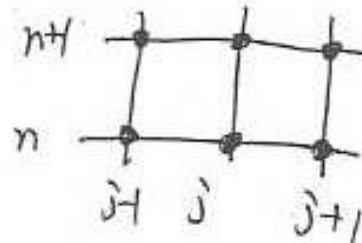
$\theta = 0$: explicit.

$0 < \theta \leq 1$: implicit, tridiagonal system, solved using Thomas algorithm.

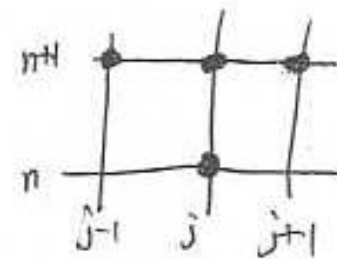
Stencil



$\theta = 0$



$0 < \theta \leq 1$



$\theta = 1$

Truncation error

$$T(x_j, t_{n+\frac{1}{2}}) = \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} - \left[(1 - \theta) \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n))}{(\Delta x)^2} \right. \\ \left. + \theta \frac{u(x_{j+1}, t_{n+1}) - 2u(x_j, t_{n+1}) + u(x_{j-1}, t_{n+1}))}{(\Delta x)^2} \right]$$

Taylor expansion at $(x_j, t_{n+\frac{1}{2}})$.

Using the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ and therefore $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2 \partial t}$ and $\frac{\partial^3 u}{\partial t^3} = \frac{\partial^2 u}{\partial x^2 \partial t^2}$, we have

$\theta \neq \frac{1}{2}$, the scheme is first order in t , second order in x .

$$\theta = \frac{1}{2}, \quad T(x_j, t_{n+\frac{1}{2}}) = -\frac{1}{12}(\Delta x)^2 \frac{\partial^4 u}{\partial x^4}(x_j, t_{n+\frac{1}{2}}) - \frac{1}{12}(\Delta t)^2 \frac{\partial^3 u}{\partial t^3}(x_j, t_{n+\frac{1}{2}}) \\ + O((\Delta t)^3) + O((\Delta x)^3).$$

Second order in both t and x .

Stability

Let

$$U_j^n = [\lambda(k)]^n e^{ik(j\Delta x)}.$$

Substitute it into the θ -method (5), we have

$$\lambda(k) = \frac{1 - 4(1 - \theta)\mu \sin^2 \frac{k\Delta x}{2}}{1 + 4\theta\mu \sin^2 \frac{k\Delta x}{2}}.$$

For this $\lambda(k)$,

$$|\lambda(k)|^n \text{ bounded, } n\Delta t \leq t_F \iff |\lambda(k)| \leq 1$$

$$\iff -1 \leq \lambda(k) \leq 1$$

$$\iff -1 \leq \frac{1 - 4(1 - \theta)\mu \sin^2 \frac{k\Delta x}{2}}{1 + 4\theta\mu \sin^2 \frac{k\Delta x}{2}} \leq 1$$

$$\iff 2(1 - 2\theta)\mu \sin^2 \frac{k\Delta x}{2} \leq 1.$$

$\frac{1}{2} \leq \theta \leq 1$, stable for all μ (unconditionally stable)

$0 \leq \theta < \frac{1}{2}$, stable if and only if $\mu \leq \frac{1}{2(1-2\theta)}$

Special cases:

$\theta = 0$, explicit method (1), $\mu \leq \frac{1}{2}$ stable

$\theta = 1$, implicit method (2), unconditionally stable

$\theta = \frac{1}{2}$, Crank-Nicolson (5), unconditionally stable

Crank-Nicolson: second order in both t and x
unconditionally stable

1.7. Lax equivalence theorem

For a well-posed linear evolutionary problem
and a consistent numerical scheme

$$\underline{\text{Stability}} \iff \underline{\text{Convergence}}$$

Well-posed problem: The solution exists, is unique, and depends continuously on the give data.